CHE 321 HW # 9 Peter Yip

1a If the stirring rate increases the reation rate will not change, which means the external diffusion is not the rate determing step. So the internal diffusion is the rate determing step

1b $C_{wp} = \eta \phi_1^2 = \frac{-r_A'(obs)\rho_c r^2}{D_e C_{AS}}$

$$C_{wp} = \left(\frac{1\times10^{-4}\ mol}{g\cdot sec}\right)\times\left(\frac{2g}{1cm^3}\right)\times\left(\frac{0.005cm}{1}\right)^2\times\left(\frac{1\ sec}{10^{-1}cm^2}\right) \times\left(\frac{1L}{1\times10^{-3}mol}\right)\times\left(\frac{10^3\ cm^3}{1L}\right)$$

$C_{wp} = 0.05$

If $C_{wp} << 1$ there is no diffusion limitations and no concentation gradient exists

1c $F_{A0}\frac{dx}{dw} = -r_A'\eta = k'C_{A0}(1-x)\eta$

$$\eta = \frac{F_{A0}}{W\cdot k'C_{A0}}[-\ln(1-x)]$$

$$\int \frac{1}{1-x}dx = \int \frac{k'C_{A0}\eta}{F_{A0}}dw$$

$$-\ln(1-x) = \frac{k'C_{A0}\eta}{F_{A0}}W$$

$$W = V \cdot \rho_B = \pi r^2 \cdot L \cdot \rho_B$$

$$W = \pi\left(\frac{10}{2}\right)^2\left(10\ m \times \frac{100\ cm}{1\ m}\right)\left(\frac{1\ g}{1\ cm^3}\right)$$

$$W = 78{,}539.7\ g$$

$$k' = \frac{r_A}{C_{A0}} = \left(\frac{1\times10^{-4}\ mol}{g \cdot sec}\right)\left(\frac{1\ L}{1\times10^{-3}\ mol}\right)$$

$$k' = 0.1\ L/g \cdot sec$$

$$\eta = \frac{1\ mol/sec}{78{,}539.7\ g \times 0.1\ L/g\,sec \times 10^{-3}\ mol/L}\ [-\ln(1-0.75)]$$

$$\eta = 0.177$$

$$\eta = \frac{1}{\phi_1} \rightarrow \phi_1 = \frac{1}{\eta} = \frac{1}{0.177}$$

$$\phi_1 = 5.665$$

$$\phi = \frac{r}{3}\sqrt{\frac{-r'_{As}\rho_c}{D_e C_{As}}} = 5.66$$

$$\frac{R}{3}(44.73) = 5.66$$

$$R = 0.38\ cm$$

2a Assum no accumulation $\frac{d}{dt} = 0$

- De is constant

$$0 = n''_A|_z \cdot \pi r^2 - n''_A|_{z+\Delta z} \cdot \pi r^2$$

$$0 = \frac{n_A''|_z}{\Delta z} - \frac{n''_A|_{z+\Delta z}}{\Delta z}$$

$$0 = -\frac{d}{dz}(n''_A)$$

$$0 = -\frac{d}{dz}\left(C D_e \frac{dx_A}{dz}\right)$$

$$0 = -D_e \frac{d}{dz}\left(\frac{dC_A}{dz}\right)$$

$$\frac{d}{dz}\left(\frac{dC_A}{dz}\right) = 0$$

$$\frac{dC_A}{dz} = A$$

$$C_A(z) = Az + B$$

@ $z = 0$ AND $C_A = C_{A,S}$

$$C_{A,S} = 0 + B \rightarrow B = C_{A,S}$$

@ $z = \frac{L}{2}$ AND $C_A = C_1$

$$C_1 = \frac{L}{2}A + C_{A,S}$$

$$2\left(\frac{C_1 - C_{AS}}{L}\right) = A$$

$$C_A(z) = \frac{2}{L}(C_1 - C_{AS}) \cdot z + C_{AS}$$

2b non-poisoning $\rightarrow$ $\eta = \frac{\tanh \phi_1}{\phi_1}$ AND $\phi_1 = L\sqrt{\frac{4k''}{dDe}}$

$L = \frac{L}{2}$ " half poision half non-poision "

$$\frac{L}{2}\sqrt{\frac{4k''}{dDe}} = \frac{\phi_1}{2} \rightarrow \eta = \frac{\tanh\left(\frac{\phi_1}{2}\right)}{\left(\frac{\phi_1}{2}\right)}$$

$$\eta (S_a \cdot k'' \, C_{A,S}) = \eta_{n-p} (S_a \cdot k'' \, C_{A,1})$$

$$\eta (\pi d L) \cdot k'' \cdot C_{AS} = \frac{\tan\left(\frac{\phi_1}{2}\right)}{\left(\frac{\phi_1}{2}\right)} \left(\pi d \frac{L}{2}\right) \cdot k'' \cdot C_1$$

$$\eta = \frac{\tan \frac{\phi_1}{2}}{2} \frac{C_1}{C_{A,S}}$$